MOLE CONCEPT

- **Atom:** The term atom was introduced by Dalton. Atom is the smallest particle of matter that takes part in a chemical reaction. Atom is also defined as the smallest particle of an element that retains all the properties of an ele ment.

- **Atomic mass unit (a.m.u.):** It is the smallest unit of mass and is used to measure the masses of atoms and subatomic particles. The mass of one a.m.u. is equal to the mass of $\frac{1}{12}$th the mass of C-12 atom. The other names of a.m.u. are Aston, Dalton and Avogram.

  Note: 1 a.m.u. = 1.66 $\times$ 10–24 g or 1.66 $\times$ 10–27 kg.

- **Atomic weight :**
  - The atomic weight or the relative atomic mass (RAM) of an element is defined as the number of times an atom of an element is heavier than the mass of $\frac{1}{12}$th of C-12 isotope's atom.
  - Relative atomic mass of an element $(\text{RAM}) = \dfrac{\text{Mass of 1 atom of that element}}{\frac{1}{12} \times (\text{Mass of C} - 12 \text{ atom})}$
  - Atomic weight has no units.
  - The relative atomic mass of an element indicates the number of times one atom of that element is heavier than $\frac{1}{12}^{\text{th}}$ of mass of C- 12 isotopes atom. For example, the atomic weight of calcium is 40. This means that an atom of calcium is on average is 40 times the mass of 1/12 the mass of C- 12 isotope's atom.
  - Atomic weights of many elements are not whole numbers due to the presence of stable isotopes.
  - The number of atoms of a particular isotope present in 100 atoms of a natural sample of that element is called its *relative abundance* which always remains constant for a given element.
  - Natural chlorine is a mixture of two isotopes with relative abundances 75% (Cl-35) and 25% (Cl-37) approximately.

    Then, the atomic weight of chlorine is $\dfrac{(75 \times 35) + (25 \times 37)}{100} = 35.5$

- Mass of one atom of an element = Relative atomic mass $\times$ mass of $\frac{1}{12}^{\text{th}}$ the mass of C- 12

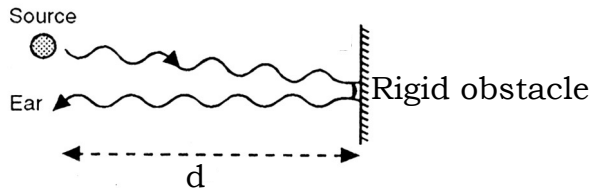
- **Molecule:** The term molecule was introduced by Avogadro. Molecule is the smallest particle of matter that exists independently and is formed by the combination of atoms. Molecule is also defined as the smallest particle of matter that can exist and retains all the properties of that substance.

  Note: A molecule splits into atoms first before taking part in a chemical reaction.

SELF TEST

1. The weight of Helium atom in grams is
   1) 2 2) 4 3) 6.64×10^{-24} 4) 1.66×10^{-24}
2. The symbol of carbon is C. It means that
   1) 'C' represents one atom of carbon.
   2) 'C' also represents 1 mole of carbon atoms.
   3) 'C' also represents 12g of C. 4) All
3. The mass of an atom of an element 'x' is 39. The number of atoms of it present in gram atomic weight of it is_____
   1) 1 2) 1.66×10^{24} 3) 6.023×10^{23} 4) 96500
4. The total mass of 100 atoms of silicon is
   1) 2800 2) 280 amu 3) $28 \times 1.66 \times 10^{-22}$g 4) 280 kg
5. The approximate number of electrons that are required to make 1 smallest unit of mass is
   1) 6.023×10^{23} 2) 1.66×10^{24} 3) 1852 4) 2500
6. The mass of one atom of an unknown element is $4 \times 1.66 \times 10^{-24}$g. The element is
   1) Hydrogen 2) Helium 3) Oxygen 4) Sulphur
7. Atoms of which of the following elements exist independently.
   1) Helium 2) Sodium 3) Calcium 4) Magnesium
8. Which of the following exist dependently?
   1) Atoms 2) Molecules 3) Ions 4)None
9. The types of forces that holds the atoms together in a molecule is
   1) Covalent bonds 2) Ionic bonds
   3) Gravitational force 4) Magnetic forces
10. Which of the following is the smallest particle of matter that exist independently?
    1) Atom 2) Molecule 3) element 4) compound
11. A: H_2O, CH_4, NH_3 ;
    B:H_2, N_2, O_2, F_2 Which of the following statement(s) is true
    a) 'A' contains homogeneous molecules.
    b) 'B' contains hetereogeneous molecules.
    c) 'A' contains hetereogeneous molecules.
    d) 'B' contains homogeneous molecules.
    1) only a,b are true 2) only c,d are true
    3) a,b,c are true d is not true 4) All are true
12. In which of the following, the smallest particles exists in diatomic form?
    1) Hydrogen gas 2) Nitrogen gas
    3) Oxygen gas 4) Flourine gas
13. The number of atoms present in a molecule is called its atomicity. Then, the atomicity of phosphorus and sulphur is_____and _____respectively.
    1) 1, 2 2) 3, 6 3) 2, 4 4) 4, 8

The repetition of sound caused by the reflection of sound waves is called an echo.



Minimum Distance of hear an Echo

The human ear can hear two sounds separately only if they reach the ear after an interval of 1/10th of a second. This is a natural feature of the human ear.

If we take the speed of sound roughly as 340 m/s, the distance travelled by sound in 1/10th of a second would be 34 m.

Thus, the distance travelled by sound in going from us (the source of sound) to the sound reflecting surface (wall), and then coming back to us should be 34 metres.

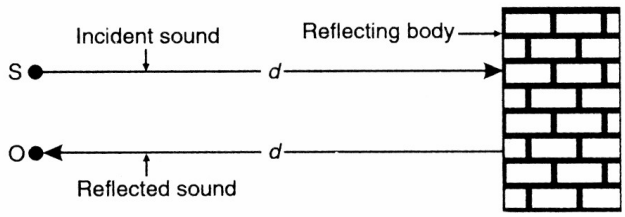
So our distance from the sound reflecting surface (like a wall, etc,) to hear an echo should be half of 34 metres which is 34/2 = 17 metres.

From this, we conclude that the minimum distance from a sound reflecting surface (like wall, etc) to hear an echo is 17 metres.

Relation between Speed of Sound, time of hearing Echo and Distance from Reflecting Body.

If ***t*** is the time in which an echo is heard, ***d*** is the distance between the source of sound and the reflecting body, and v is the speed of sound, then the total distance travelled by the sound is ***2d***.

S = Source of sound d = Distance between source of sound and reflecting body.
O = Observer



$\therefore$ In the time t seconds, distance travelled by sound = 2d

Hence, in time 1 seconds, distance travelled by sound $= \frac{2d}{t}$

But, distance travelled by sound in 1 second = Speed of sound

$\therefore \quad v = \frac{2d}{t}$ or $t = \frac{2d}{v}$ or $d = \frac{v \times t}{2}$

11. The gravitational force between two bodies is $6.67\times10^{-7}N$ when the distance between their centres is 10m. If the mass of first body is 800 kg, then the mass of second body is
1) 1000 kg 2) 1250 kg 3) 1500 kg 4) 2000 kg

12. Two identical spheres each of radius R are placed with their centres at a distance nR, where n is integer greater than 2. The gravitational force between them will be proportional to
1) $1/R^4$ 2) $1/R^2$ 3) R^2 4) R^4

13. A satellite is orbiting round the earth. If both gravitational force and centripetal force on the satellite is F, then, net force acting on the satellite to revolve round the earth is
1) F/2 2)F 3)2F 4) Zero

14. Mass M=1 unit is divided into two parts 'x' and (1– x). For a given separation the value of 'x' for which the gravitational force between them becomes maximum is
1) 1/2 2) 3/5 3) 1 4) 2

15. Imagine a new planet having the same density as that of earth but it is 3 times bigger than the earth in size. If the acceleration due to gravity on the surface of earth is g and that on the surface of the new planet is g', then
1) $g' = 2g$ 2) $g' = 3g$ 3) $g' = 4g$ 4) $g' = 5g$

16. Mass of the earth is 81 times the mass of the moon and distance between the earth and moon is 60 times the radius of the earth. If R is the radius of earth, then the distance between the moon and the point on the line joining the moon and earth where the gravitational force becomes zero is
1) 30 R 2) 15 R 3) 6R 4) 9R

17. The gravitational force of attraction between two bodies is F newtons. If the mass of each body and the distance between them are doubled, then the gravitational force between them in newton is
1) 16F 2) $\frac{F}{16}$ 3) $\frac{F}{4}$ 4) F

18. If 'R' is radius of the earth, the height above the surface of earth where the weight of a body is 36% less than its weight on the surface of the earth is
1) $\frac{4R}{5}$ 2) $\frac{R}{5}$ 3) $\frac{R}{6}$ 4) $\frac{R}{4}$

19. The depth from the surface of the earth at which the acceleration due to gravity will be 75% of the value on the surface of the earth (Radius of earth = R)
1) $\frac{R}{4}$ 2) $\frac{3R}{4}$ 3) $\frac{R}{2}$ 4) $\frac{R}{8}$

20. The acceleration due to gravity at a distance of $\frac{R}{4}$ from the centre of the earth (Assume acceleration due to gravity at the surface of the earth is g)
1) 2g 2) 4g 3) $\frac{g}{2}$ 4) $\frac{g}{4}$

21. The weight of a body at a distance 2R from the centre of the earth of radius 'R' is 2.5 N. If the distance is 3R from the centre of the earth, the weight of the body is
1) 1.1 N 2) 2.1 N 3) 3.1 N 4) 4.1 N

21. A ball of mass 50 kg is moving horizontally with a velocity of 20 m/s. A man hits the ball and the ball travels back in the opposite direction with same velocity. Find the change in momentum. If the ball is contact with the bat for a time of 0.1 s, find the force exerted by the bat on the ball

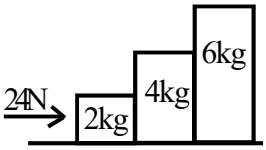
1)50,000N 2)10,000N 3) 20,000N 4)30,000N

22. A certain force applied to mass m_1 gives it an acceleration of $6ms^{-2}$.The same force applied to mass m_2 gives it an acceleration of $3ms^{-2}$. If the two masses are fixed together and the same force is applied to the combination, acceleration of it would be

1) $9ms^{-2}$ 2) $4.5ms^{-2}$ 3) $3\ ms^{-2}$ 4) $2ms^{-2}$

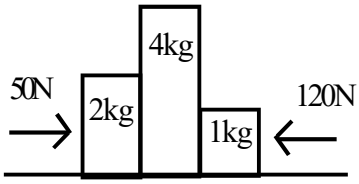
23. Three blocks of masses, 2 kg, 4 kg, 6 kg are in contact with each other as shown in the figure, if a force of 24 N is applied from left side, then the acceleration of the system is

1) 4 m/sec^2
2) 2 m/sec^2
3) 1 m/sec^2
4) 8 m/sec^2



24. Three blocks of masses 1 kg, 4 kg and 2 kg are placed on a smooth horizontal surface as shown in the figure. Two horizontal forces 120 N and 50 N are applied on the system. The acceleration of the system is

1) 17.5 ms^{-2}
2) 10 ms^{-2}
3) 24.3 ms^{-2}
4) 70 ms^{-2}

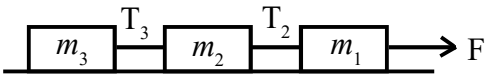


25. When a horizontal force of 200 N is applied on a body, the acceleration produced in it is $1.5m/s^2$. When the force is 300 N, the acceleration produced in the same body is $2.5m/s^2$. Then the mass of the body is

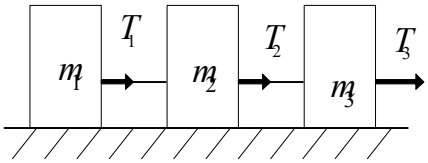
1) 50 kg 2) 25 kg 3) 100 kg 4)125 kg

26. Three blocks of masses m_1 = 1 kg , m_2 = 2kg and m_3 = 3 kg are connected as shown in the figure. A force of 12 N applied on the mass 'm_1' as shown. Then the acceleration of the system is

1) 12 ms^{-2} 2) 6 ms^{-2}
3) 4 ms^{-2} 4) 2 ms^{-2}



27. Three blocks of masses m_1 , m_2 and m_3 are connected by a massless string as shown in figure on a frictionless table. They are pulled with a force T_3 = 40 N. If $m_1 = 10kg$, $m_2 = 6kg$ and $m_3 = 4kg$, then tension T_2 will be



1)10 N 2) 20 N 3) 32 N 4) 40 N

14. A stone is dropped from the top of a tower of height 125 m. The distance travelled by it during last second of its fall is (g = 10 ms^{-2})
1) 22.5 m 2) 45 m 3) 75 m 4) 90 m

15. The ratio of the times taken by freely falling body to cover first metre, second metre and soon is
1) $\sqrt{1}:\sqrt{2}:\sqrt{3}$ 2) $\sqrt{1}:\sqrt{2}-\sqrt{1}:\sqrt{3}-\sqrt{2}$ 3) $\sqrt{2}:\sqrt{4}:\sqrt{8}$ 4) 2 : 3 : 4

16. If the distance travelled by a freely falling body in the last second of its journey is equal to the distance travelled in the first two seconds, the time descent of the body is
1) 5 s 2) 1.5 s 3) 2.5 s 4) 3 s

17. A freely falling body gains a velocity v in travelling a distance 'h'. The distance to be travelled after this point to double its velocity is
1) 4 h 2) 2 h 3) 3 h 4) h

18. A stone falls from a height H. The earth stops to attract it when it has fallen through $\frac{H}{2}$. The stone reaches the ground with a velocity of
1) $\sqrt{2gH}$ 2) $\sqrt{\frac{gH}{2}}$ 3) $\frac{\sqrt{gH}}{2}$ 4) $\sqrt{gH}$

19. Drops of water fall from the roof of a building 9 m high at regular intervals of time, the first drop reach the ground at the same instant fourth drop starts its fall. What are the distances of the second and third drops from the roof?,
1) 4 m and 1 m 2) 4 m and 2 m
3) 6 m and 2 m 4) 6 m and 3 m

Projected vertically upwards

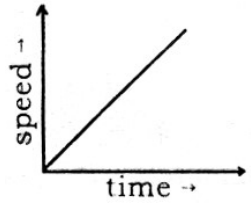
20. A body is projected with a velocity 50 ms^{-1}. What is its velocity after 2 seconds (g = 10m/s^2)
1)70m/s 2)50m/s 3)40m/s 4)30m/s

21. A body is projected with a velocity 30 ms^{-1}. What is distance travelled by it in 4 seconds?(g = 10m/s^2)
1)45m 2)50m 3)40m 4)30m

22. A body is projected with a velocity 30 ms^{-1}. What is displacement made by it with in 4 seconds?(g = 10m/s^2)
1)45m 2)50m 3)40m 4)30m

23. A body is projected with a velocity 50 ms^{-1}. Distance travelled in 6th second is (g =10ms^{-2})
1) 5 m 2) 10 m 3) 15 m 4) 20 m

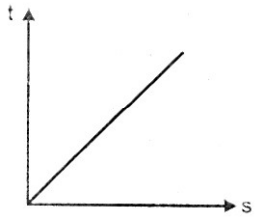
24. A body is projected vertically upwards with a velocity of 9.8 m/s. What time it reach the point of projection (g 9.8m/.s^2)
1) 0.5 sec 2) 1 sec 3) 2 sec 4) 1.5 sec

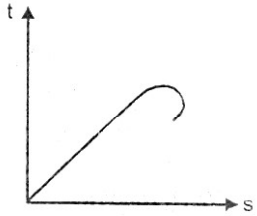


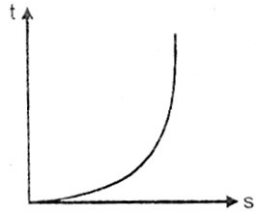
1) Speed is constant 2) Speed is decreasing

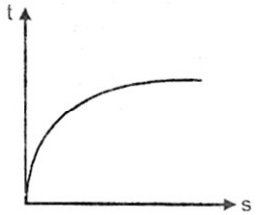
3) Speed is increasing 4) Speed increases to a maximum and then decreases

7. Which of the following time-displacement graph is not possible in nature?

1) 

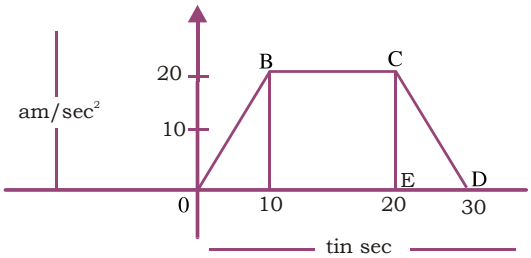
2) 

3) 

4) 

Q.No : 8 to 10

Figure shows the time acceleration graph for a particle in rectilinear motion.



8. Area of acceleration-time graph gives

1) speed 2)acceleration

3)change in velocity 4) distance

9. Area under OBCEO is

1)300 units 2)400 units 3)500 units 4)600 units

10. The average acceleration in first twenty sec is

1)1 m/s^2 2)5 m/s^2 3)15 m/s^2 4)3 m/s^2

Newtons second Law of Motion

INTRODUCTION

- **What is force**? How is it related with laws of motion? To the answer of the first question, let us know, what happens when something that we call force acts on any body?
- Force is something that can do the following:
  - (i) Can deform a body.
  - (ii) Can make a static body moving.
  - (iii) Can stop a moving body.
  - (iv) Force can make a moving object move faster.
  - (v) Force can change direction of moving objects.
- Thus we find that force is something that is the cause of motion.
- In this chapter, we shall study the motion of a particle along with the causes of such motion. Rigid bodies undergoing translation may also be treated as particles for the analysis of their translational motion.

  We shall learn how does force govern any motion with help of Newton's laws of motion.
- **Newton's First law of motion**

  Every body continues to be in a state of rest or uniform motion in a straight line, except observation externall force acts on the body under observation.
- **Inertia**

  From Newton's first law of motion, it is clear that a body is unable to change its state of rest or motion by itself. This property of the body is known as inertia.

  Inertia of a body may be defined as the tendency of a body to oppose any change in its state of rest or uniform motion.

  For example, a book lying on a table will remain where it is placed unless it is displaced. Similarly, a ball rolling on a horizontal surface keeps on rolling unless the force of friction between the ball and the surface stops it.

  Mass and inertia : Inertia is an inherent property of each body by virtue of which it has a tendency to resist the change in its state of rest or state of uniform motion. The property of inertia is due to the mass of a body. The greater the mass, the greater is the inertia of a body.

  For example, if a body has a mass of 1kg and another body has a mass of 20 kg, then the body having 20 kg mass will have more inertia since its mass is more.
- **Types of inertia**

  Inertia can be divided into three types.

  i) Inertia of rest ii) Inertia of motion

  iii) Inertia of direction

8. If the same force acts on two bodies of different masses, then the accelerations produced are in the inverse ratio of their masses

$\frac{a_1}{a_2} = \frac{m_2}{m_1}$.

9. The ratio of times for which the same force must act on two bodies of different masses initially at rest to have

i) the same displacement is $\frac{t_1}{t_2} = \sqrt{\frac{m_1}{m_2}}$

ii) the same final velocity is $\frac{t_1}{t_2} = \frac{m_1}{m_2}$

iii) the same final momentum is $\frac{t_1}{t_2} = \frac{1}{1}$.

10. Let a force gives an acceleration a_1 for a body and gives an acceleration a_2 for another body. If two bodies are connected together and the same force is applied, then acceleration produced in the system is $a = \frac{a_1 a_2}{a_1 + a_2}$.

11. **Working with newton's laws**

The main steps are :

(i) Draw the free-body diagram of each body, showing all the forces acting on it.

(ii) If the bodies are directly connected with the same in extensible string and moving along the same direction, they will have acceleration of same magnitude. Identity the direction of acceleration for each body.

(iii) Taking one axis along the direction of acceleration and the other perpendicular to it. Then resolve the forces into the respective components.

(iv) Form two equations from each free-body diagram :
Net force in the direction of acceleration = ma
Net force perpendicular to the direction of the acceleration = 0.
Force which are perpendicular to the direction of acceleration must balance each other.

12. If a ball hits a wall normally with a velocitiy V and rebounds with same velocity

Change in momentum = 2mv

Force acting on the body = $\frac{2mv}{t}$,Where 't' is the time of collision.

13. When a machine gun fires 'n' bullets in a time t then the force required to hold the gun is $F = \left(\frac{mnv}{t}\right)$

14. When a metallic plate of mass M is held in mid air by firing n shots per second, if the bullets stop dead after striking the plate then mnu = Mg.
If the shots rebound with same velocity 2mnu = Mg.

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